Year 9 Science – Booklet 13: Physics

Space Physics (Motion on Earth and in Space) — ANSWERS

Quick Test (p.89)

1. 1 N upward (an equal and opposite reaction force, since the apple isn't moving).
2. The gradient (steepness) of the line tells us the object's speed – steeper = faster, shallower = slower, and a horizontal line means the object is stationary.
3. They're moving in opposite directions, so add the speeds: 70 + 60 = **130 mph**.
4. The slope (gradient) tells you the object's speed.

Quick Test (p.91)

1. 12 hours (day and night are equal length everywhere at an equinox).
2. The Earth's axis is tilted, so as it orbits the Sun, the Northern and Southern Hemispheres receive different amounts of solar radiation at different times of year.
3. mass = W/g = 800/10 = 80 kg. Weight on Jupiter = 80 × 25 = **2000 N**.
4. Mercury has a much smaller mass than Earth, and a smaller mass produces a weaker gravitational pull, so g is much lower.

Vocabulary Builder (15 marks)

1. Complete the sentences (3 marks)

a) The Moon is in orbit around the **Earth**. b) The Earth is in orbit around the **Sun**. c) The Sun is in orbit around the centre of the **galaxy**.

2. Box stationary on a table

a). Gravity (weight), acting downward.

b). The two forces (upward contact force and downward weight) are equal in size and opposite in direction, so they balance/cancel out and the box stays stationary.

3. Equilibrium or not, with explanation (6 marks)

Car (accelerating): **not** in equilibrium – accelerating means the forces are unbalanced.
Cyclist (constant speed): **in** equilibrium – constant speed means the forces are balanced.
Bus (decelerating): **not** in equilibrium – decelerating means the forces are unbalanced.

4. Best unit for distance to other galaxies

Tick **light-year**.

5. What keeps Jupiter's moons in orbit?

Gravity (Jupiter's gravitational field).

6. Which statement about the Earth is incorrect?

Tick **"The Earth rotates on its axis from east to west."** (it actually rotates west to east, which is why the Sun appears to rise in the east).

Maths Skills (28 marks)

1. Weight, mass and gravitational field strength

a). W = mg = 80 × 10 = **800 N**

b) i). Tick **m = W/g**

b) ii). m = W/g = 4/10 = **0.4 kg**

b) iii). W = mg = 0.4 × 1.6 = **0.64 N**

2. Speed, distance and time

a) i). 1.5 km = **1500 m**

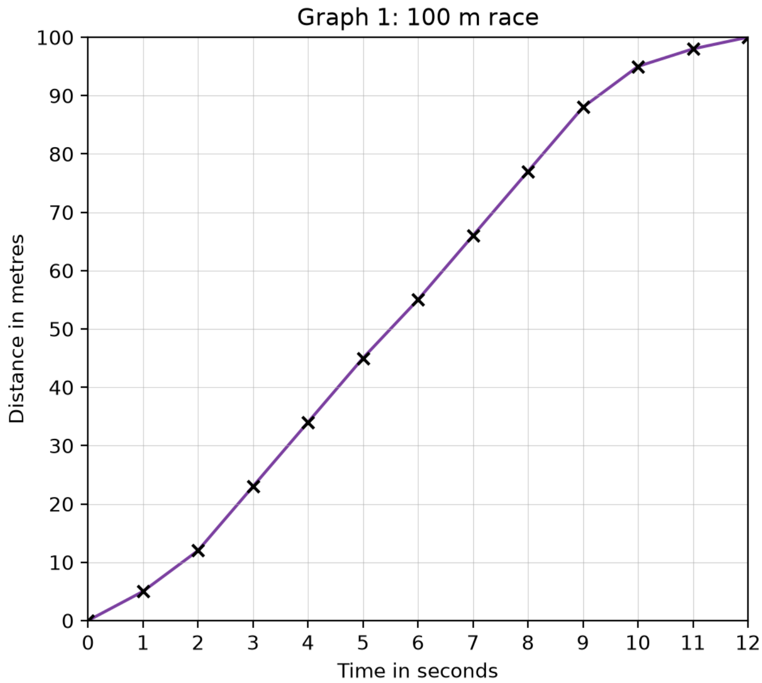
a) ii). 2.5 min = **150 s**

a) iii). speed = 1500/150 = **10 m/s**

b) i). Tick **time = distance/speed**

b) ii). time = 8000 m ÷ 3.2 m/s = **2500 s** = 2500/60 ≈ **41.7 minutes**

3. Athlete's 100 m race



c). average speed = 100 m ÷ 12 s = **8.33 m/s** (3 s.f.)

d) Distances travelled in each interval

i). 0–1 s: 5 – 0 = **5 m**

ii). 1–2 s: 12 – 5 = **7 m**

iii). 6–7 s: 66 – 55 = **11 m**

iv). 11–12 s: 100 – 98 = **2 m**

e) Using the distances above

i). First 3 seconds (5 m, 7 m, 11 m per second) – tick **increases**.

ii). Last 3 seconds (7 m, 3 m, 2 m per second) – tick **decreases**.

iii). Between 6–9 s (11 m, 11 m, 11 m per second) – tick **constant**.

f) True or false

i) Steeper slope = increasing speed – **True**
ii) Less steep slope = decreasing speed – **True**
iii) Unchanging slope = constant speed – **True**
iv) Less steep slope = increasing speed – **False** (a less steep slope means the speed is decreasing, not increasing)

Testing Understanding (27 marks)

1. Spring and mass

a) i). Gravity (weight).

a) ii). The two forces are balanced (equal size, opposite direction) – the mass is in equilibrium.

b). Tick **"The upward force is greater than the downward force."** (it accelerates upward, so there must be a net upward force).

2. Delivery van

a) i). unbalanced force = 2200 – (400+300+300) = **1200 N**, direction = **forwards**.

a) ii). Tick **van speeds up**.

3. Resultant force table

Object	Driving (N)	Friction (N)	Air res. (N)	Resultant (N)	Direction
cyclist	0	5	15	20	reverse
car	900	400	500	0	none
bus	3000	600	700	1700	forward

b) i) Cyclist: resultant force acts backwards, so the cyclist **slows down**.
ii) Car: zero resultant force, so it travels at a **constant speed**.
iii) Bus: resultant force acts forwards, so it **speeds up**.

4. Distance-time sketch graphs A, B, C, D

a). Stationary at some stage: **A and B**.

b). Accelerating for the whole motion: **D**.

c). Reverses at some stage: **B**.

d). Constant, brief acceleration, then constant higher speed: **A**.

5. Cyclists Z and Y

a). Speed of Z = 40 m ÷ 5 s = **8 m/s**.

b). Speed of Y = 25 m ÷ 5 s = **5 m/s**.

c). Relative speed (same direction, so subtract) = 8 – 5 = **3 m/s**.

d). At t = 10 s: Z has travelled 8 × 10 = 80 m, Y has travelled 5 × 10 = 50 m, so they are 80 – 50 = **30 m** apart. (Equally: relative speed 3 m/s × 10 s = 30 m.)

6. Tilted Earth

a). The Earth spinning (rotating) on its axis once every 24 hours, so different parts face towards or away from the Sun at different times.

b). The Earth's axis is tilted, so as it orbits the Sun over the year, the UK (Northern Hemisphere) is sometimes tilted towards the Sun (more direct sunlight, longer days = summer) and sometimes tilted away (less direct sunlight, shorter days = winter).

7. The Milky Way (diameter ~106 000 light-years)

a). A light-year is the distance that light travels in one (Earth) year.

b). Because distances in space are so enormous that kilometres or miles would need extremely large, unwieldy numbers; the light-year is a far more manageable/convenient unit for such vast distances.

Working Scientifically (16 marks)

1. Equilibrium

a). A stationary object is in equilibrium if the forces on it are balanced (no resultant force), so it stays at rest.

b). A moving object is in equilibrium if the forces on it are balanced (no resultant force), so it keeps moving at a constant speed in a constant direction, without speeding up, slowing down or changing direction.

c) Antonio's metre rule (four vertical forces)

i). newton-meter 1 (1.9 N): **up**. Weight of metre rule (1.0 N): **down**. Weight of the load (5.0 N): **down**. newton-meter 2 (4.1 N): **up**.

ii). Total upward force = 1.9 + 4.1 = 6.0 N. Total downward force = 1.0 + 5.0 = 6.0 N. Since these are equal, the forces are balanced, confirming the metre rule is in equilibrium.

d) Shari's setup (6.0 N load)

i). She could adjust the height of the two newton-meters and use a spirit level on the metre rule, checking the bubble is centred, until the rule reads level/horizontal.

ii). Total downward force = 1.0 (rule) + 6.0 (load) = 7.0 N. For equilibrium, total upward force must also = 7.0 N. Newton-meter 2 reading = 7.0 – 2.2 = **4.8 N**.

Science in Use (7 marks)

1. Voyager 1 & 2 slingshots

a) i). A gravity assist manoeuvre.

a) ii). Voyager 1: **2** slingshots (Jupiter and Saturn).

a) iii). Voyager 2: **4** slingshots (Jupiter, Saturn, Uranus and Neptune).

b) i). Jupiter.

b) ii). Without the gravity-assist boosts, the spacecraft wouldn't have gained the extra speed needed to escape the Sun's gravity. With no fuel left to power themselves further, the Sun's gravitational pull would have kept slowing them down, and they would never have escaped the Solar System.

c). The Milky Way.